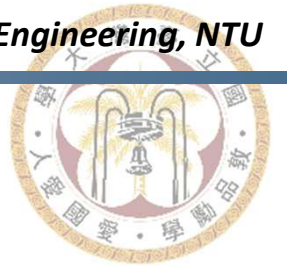
Advanced Computer-Aided VLSI System Design

Introduction

Graduate Institute of Electronics Engineering, National Taiwan University



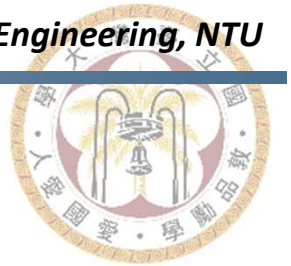
NTU GIEE



Introduction

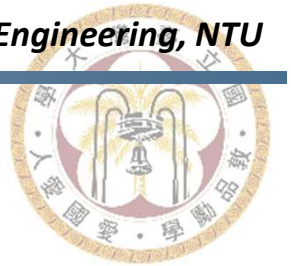
- Objective
  - Students aim to learn how to design SoC and digital circuits following a **VLSI system design flow in a FinFET technology. Electronic design automation (EDA) tools** will be used extensively in the semester

- Course content
  - Introduction to FinFET
  - Advanced logic synthesis
  - Low power design
  - System bus and protocol
  - Clock domain crossing
  - Design for testability
  - APR with FinFET standard cells
  - Sign-off



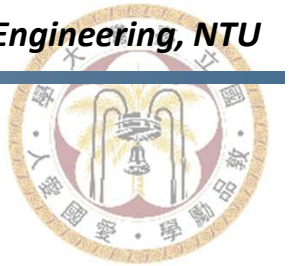
Course Schedule (1)

Week	Date	Lecture	Lecturer	Place	Lab	HW assign	HW due
01	2/24	Introduction	楊家驤教授 蔡岳峰	MD 231			
02	3/03	FinFET Technology	胡璧合教授	Online			
03	3/10	Logic Synthesis & Low Power	張均豪	Online	Lab0		
04	3/17	Logic Synthesis & Low Power	張均豪	Online	Lab1	HW1	
05	3/24	Multi-Clock & PLL	陳柏任	Online	Lab2		
06	3/31	System Bus	陳柏任	Online		HW2	HW1
07	4/07	Design for Testability	吳燦霖	Online	Lab3	**Midterm project**	
08	4/14	Midterm Project Q&A Final Project Announcement	蔡岳峰	Online		**Final project** /HW3	HW2



Course Schedule (2)

Week	Date	Lecture	Lecturer	Place	Lab	HW assign	HW due
09	4/21	APR	黃佳穎	Online	Lab4		
10	4/28	APR	黃佳穎	Online	Lab5		HW3
11	5/05	Final Project Review	TAs	TBD		HW4	
12	5/12	Final Project Review	TAs	TBD			**Midterm project**
13	5/19	Timing Sign-Off	蔡岳峰	Online	Lab6		
14	5/26	Physical Sign-Off	蔡岳峰	Online	Lab7		HW4
15	6/02	Invited Talk	TBD	Online			
16	6/09	Invited Talk	TBD	Online			**Final project**
17	6/16	Final Project Presentation		TBD			



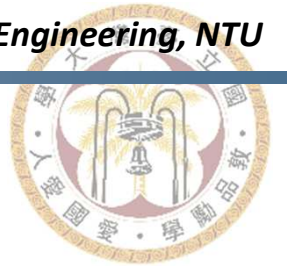
Online Course Link

- Webex:

  https://nationaltaiwanuniversity-581534.my.webex.com/nationaltaiwanuniversity-581534.my/j.php?MTID=me621bf8817a65ebda9823d27cb44f1ba

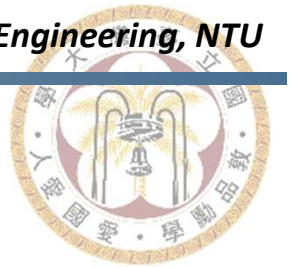
- Please log in using "[student ID]-[Name]" for TA to verify your identity.

- Your microphone is set to be muted by default. If you have any questions, please use the chatroom to ask.



TA Contact Information

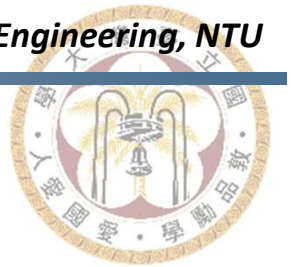
Lead TA	**蔡岳峰**	**f12943014@ntu.edu.tw**
Midterm Project	邱嚴盛	r14943004@ntu.edu.tw
Lecture: Logic Synthesis and Low Power	張均豪	r12k41025@ntu.edu.tw
Lecture: Multi-clock Design and Bus	陳柏任	d13943013@ntu.edu.tw
Lecture: Design for Testability	吳璨霖	r13943017@ntu.edu.tw
Lecture: APR	黃佳穎	r13943003@ntu.edu.tw
Lecture: Sign-Off	蔡岳峰	f12943014@ntu.edu.tw
Homework 1	高珩翔	r14943009@ntu.edu.tw
Homework 2	江承恩	r13943008@ntu.edu.tw
Homework 3	徐以帆	r13943005@ntu.edu.tw
Homework 4	張禾牧	r14943011@ntu.edu.tw



Course Information

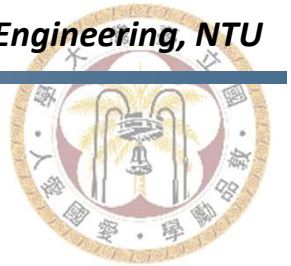
- Course materials
  - Lecture notes
  - Technical documents

- Course website
  - https://cool.ntu.edu.tw/courses/56286



Grading Policy

- **Participation (Lab): 5%**
- **Homework: 30%**
  - Submitted to NTU COOL
  - Deadline: Tuesday afternoon (13:59 pm)
- **Midterm Project: 30%**
  - A team of 1 student
- **Final Project: 35%**
  - A team of 2 students
- **No late submissions (unless for special cases). Plagiarism is prohibited.**



Enrollment Announcement

- You should pass CVSD before taking Advanced CVSD
- This course is **not** open for auditing

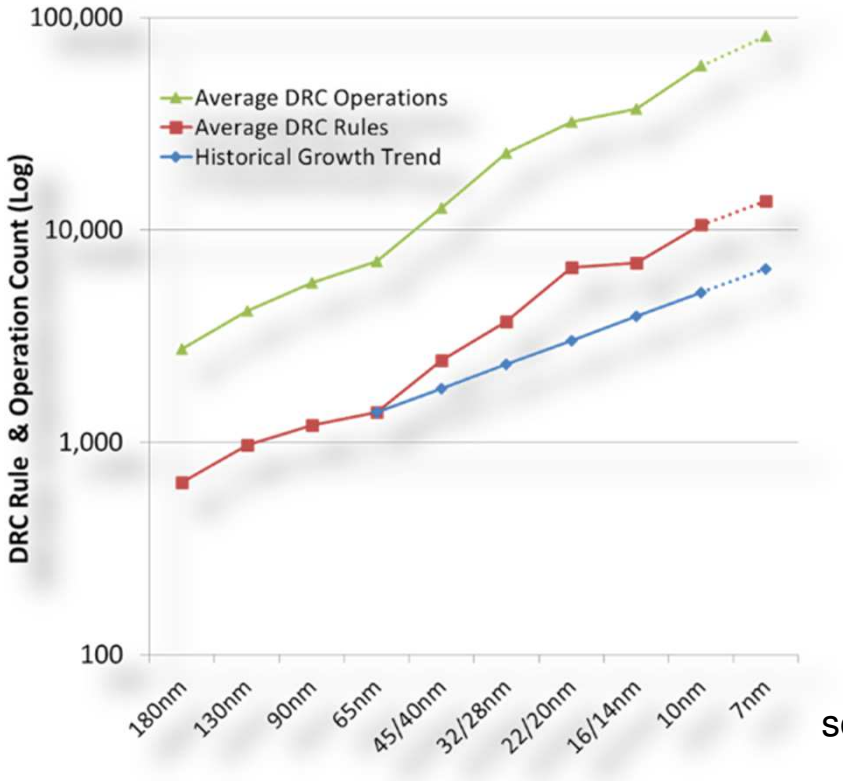
Course Registration Form

- Form: https://forms.gle/TZxW9nvdeKLqJcSn8
- **Deadline: 2/28 23:59**

- Ensure you complete these forms to complete the registration
  - **Sign-In Sheet (簽到表)**
  - **Personal Data Collection Agreement (個人資料同意書)**
  - **Interview Form (訪談表)**
  - **Honor and Code licensing Agreement (榮譽守則暨程式碼授權同意書)**

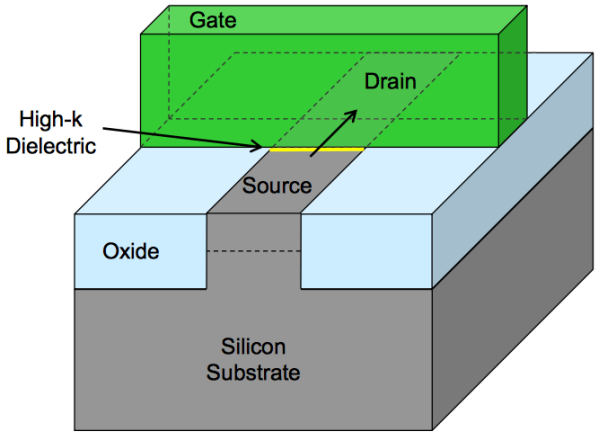
TSMC ADFP

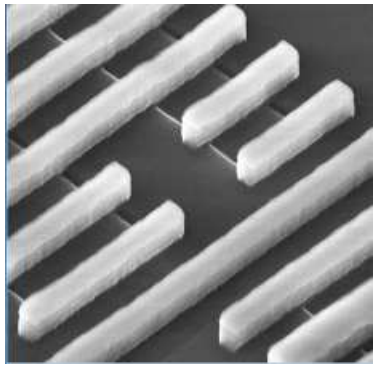
- Academic Design Forster Package (ADFP) is a virtual 16nm FinFET CMOS technology developed by TSMC for teaching purposes only
- ADFP supports full-custom and standard-cell design
- ADFP contains much more design rules than planar technology



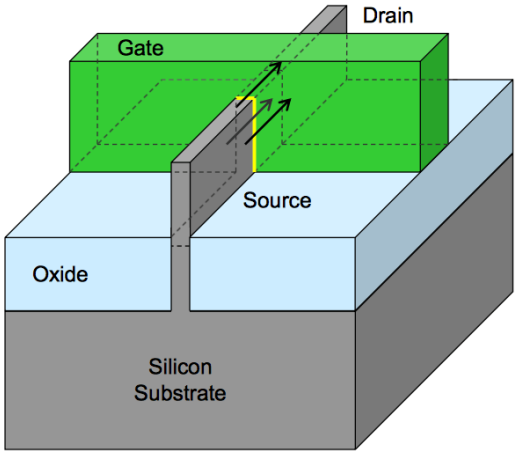
source: Tech Design Forum

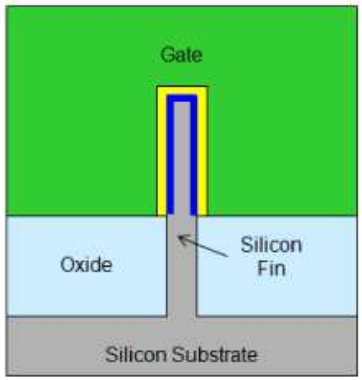
FinFET



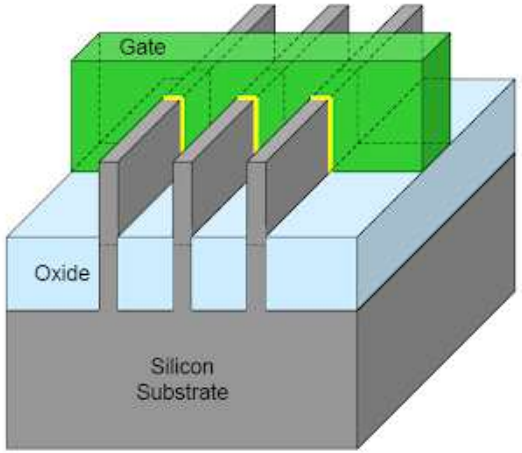


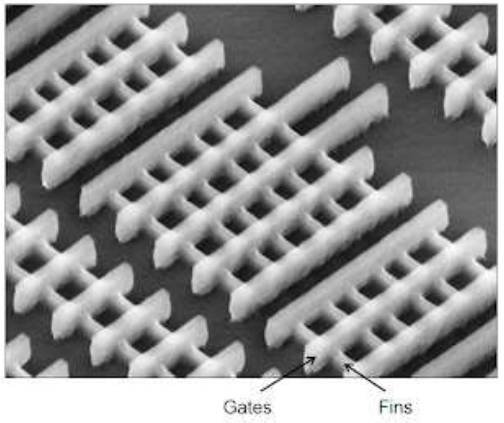
Planar Transistor





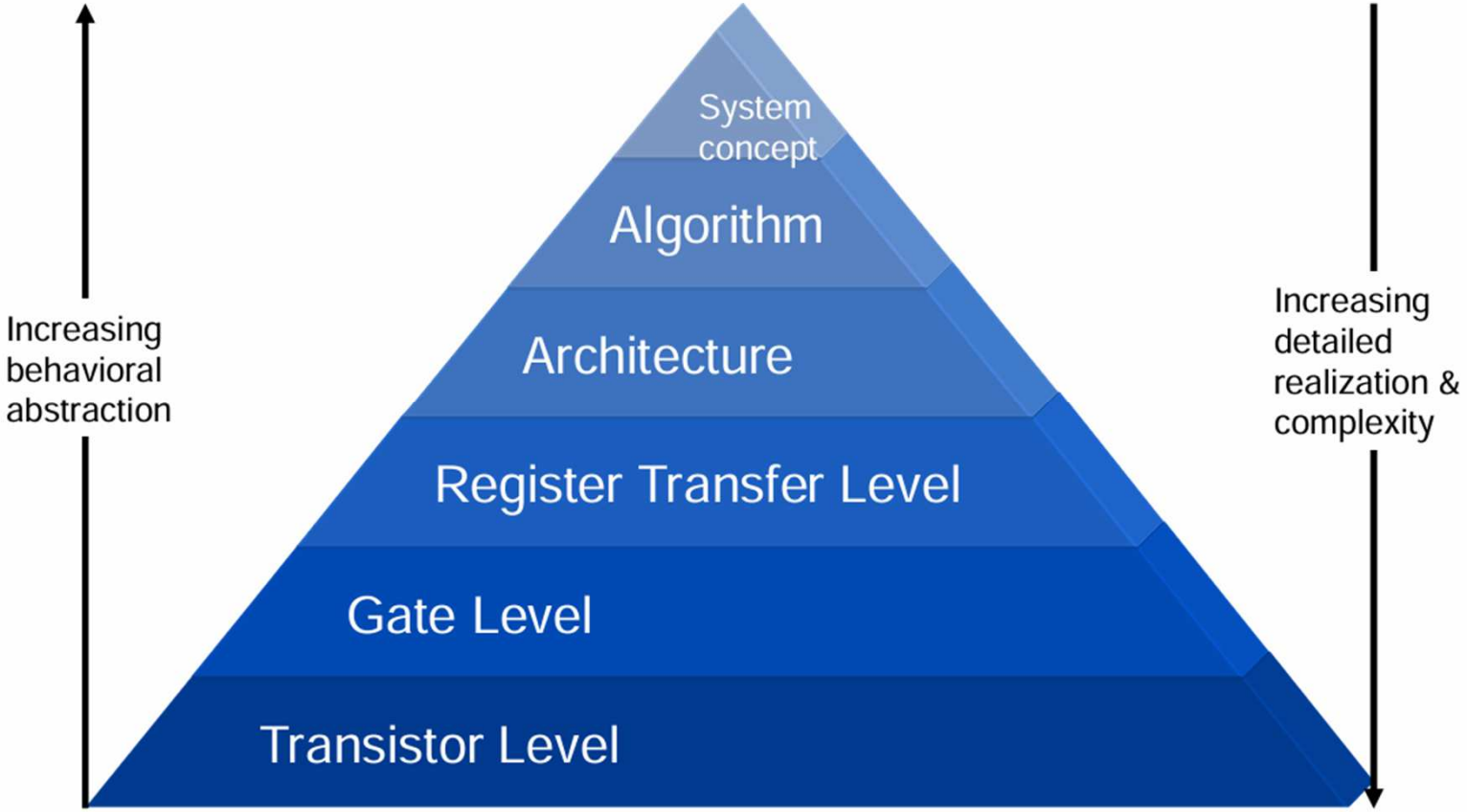
FinFET Transistor

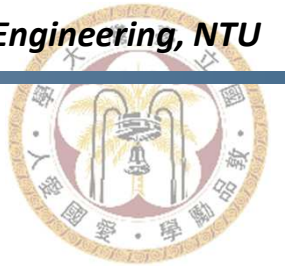




Multi-Fin Transistor

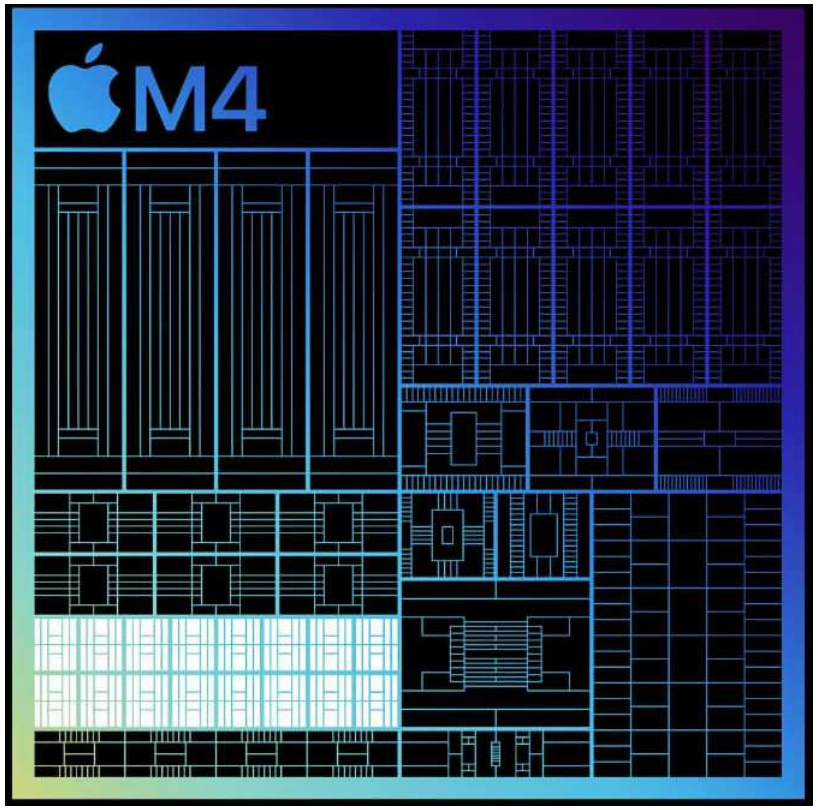
System Integration





SoC Design

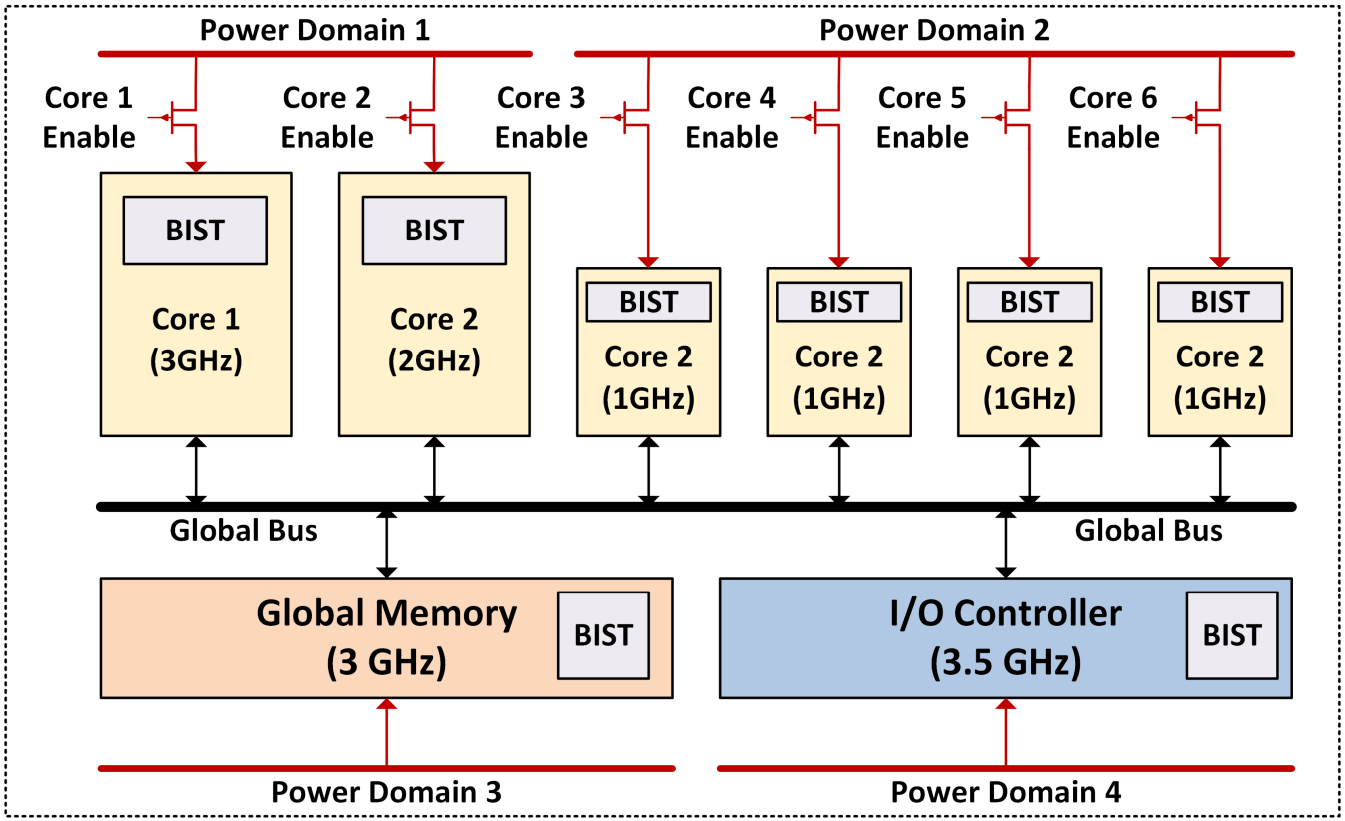
- Modern SoCs integrate CPU, GPU, NPU, and other modules
  - Apple M4 chip contains 28B transistors and only consumes 22 watt



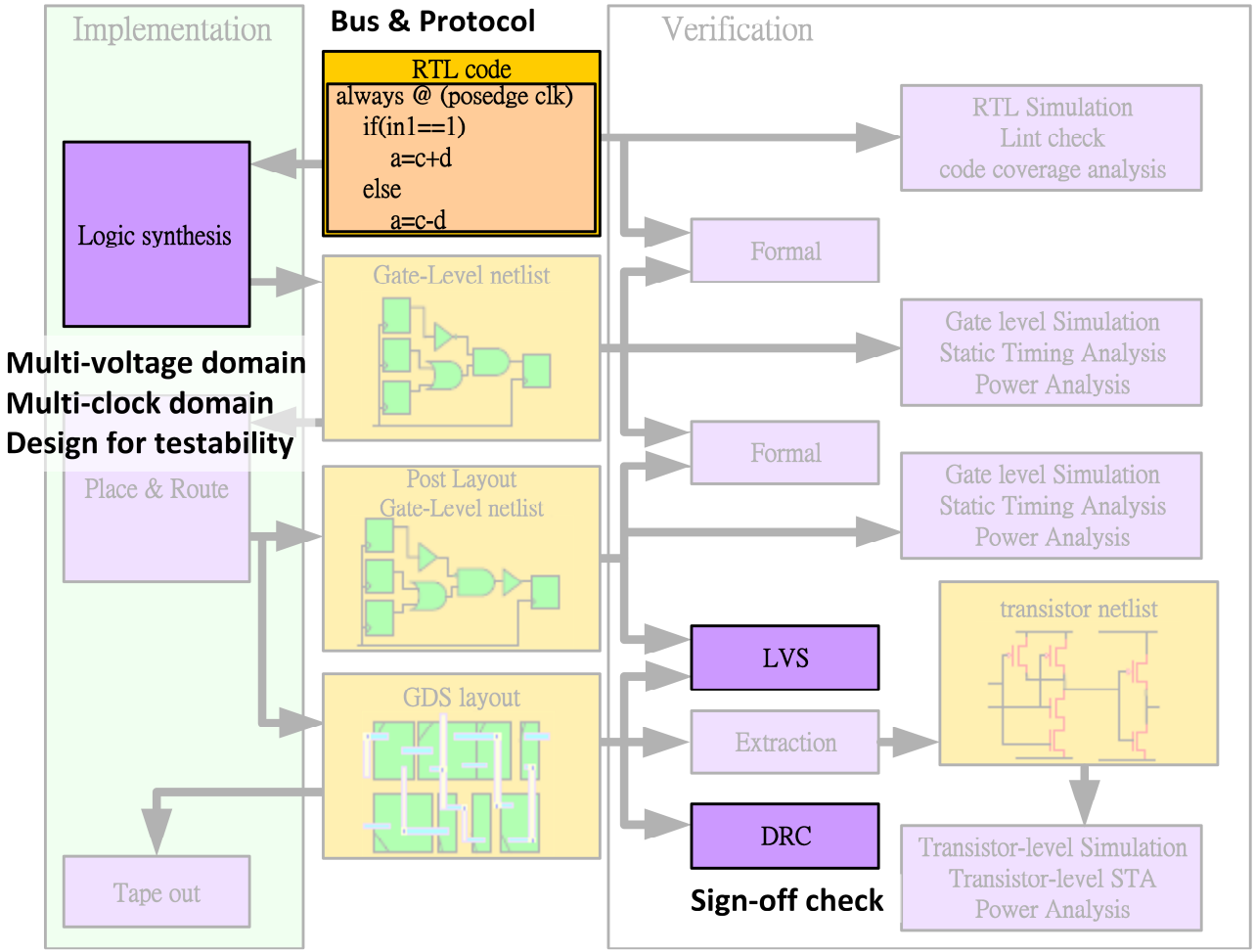
Source: Apple

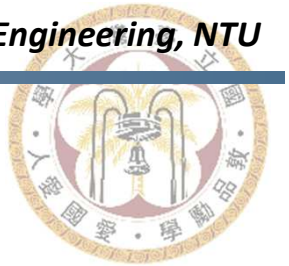
SoC Design

- System integration: Bus, Protocol
- Power reduction: Multi-voltage domain, Multi-clock domain
- Testability and reliability: Design for testability, Sign-off check



Cell-Based VLSI Design





System Bus

- Configure units connected to the system bus
- Appropriate protocols reduce the transmission latency and increase the utilization of the circuit (example: burst mode)
- This course covers **AXI, AHB**, and **APB**

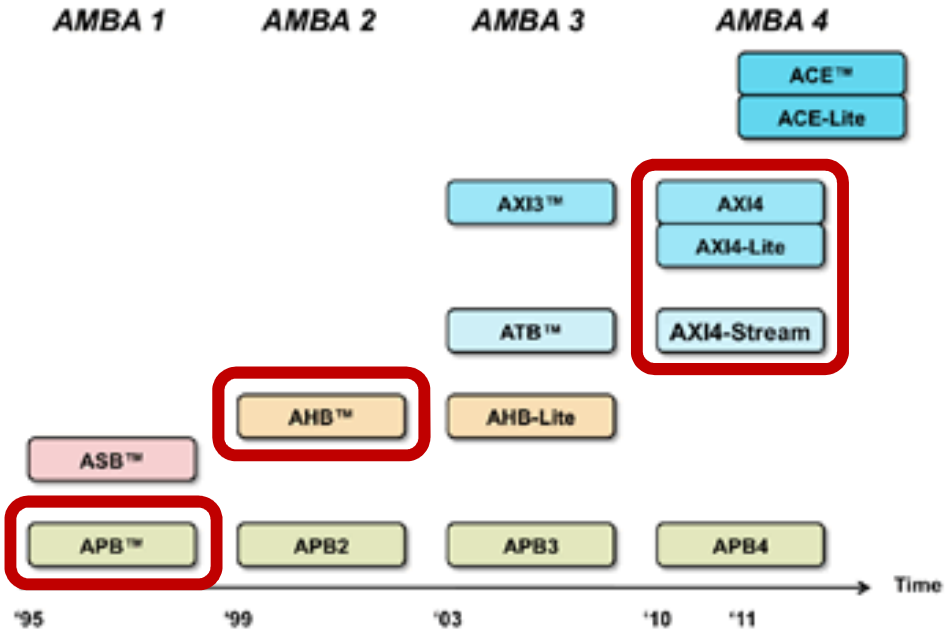
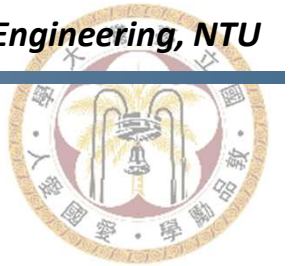
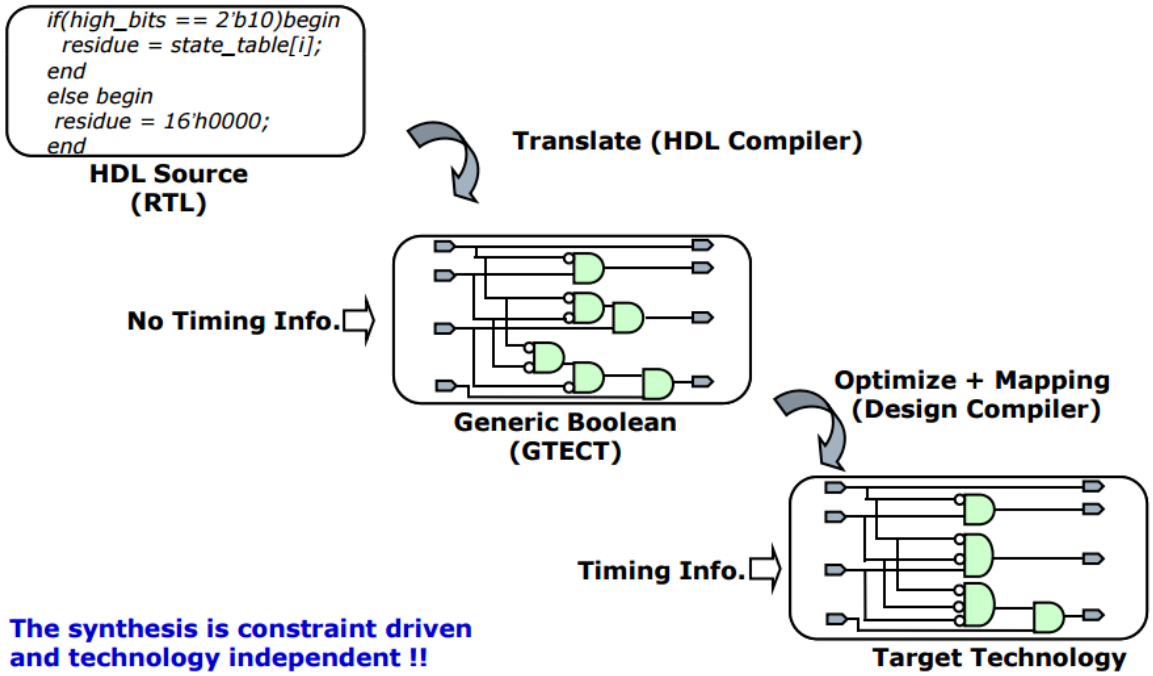


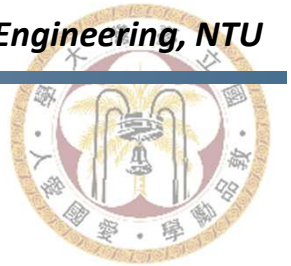
Figure 1 – Evolution of AMBA Standards



Logic Synthesis

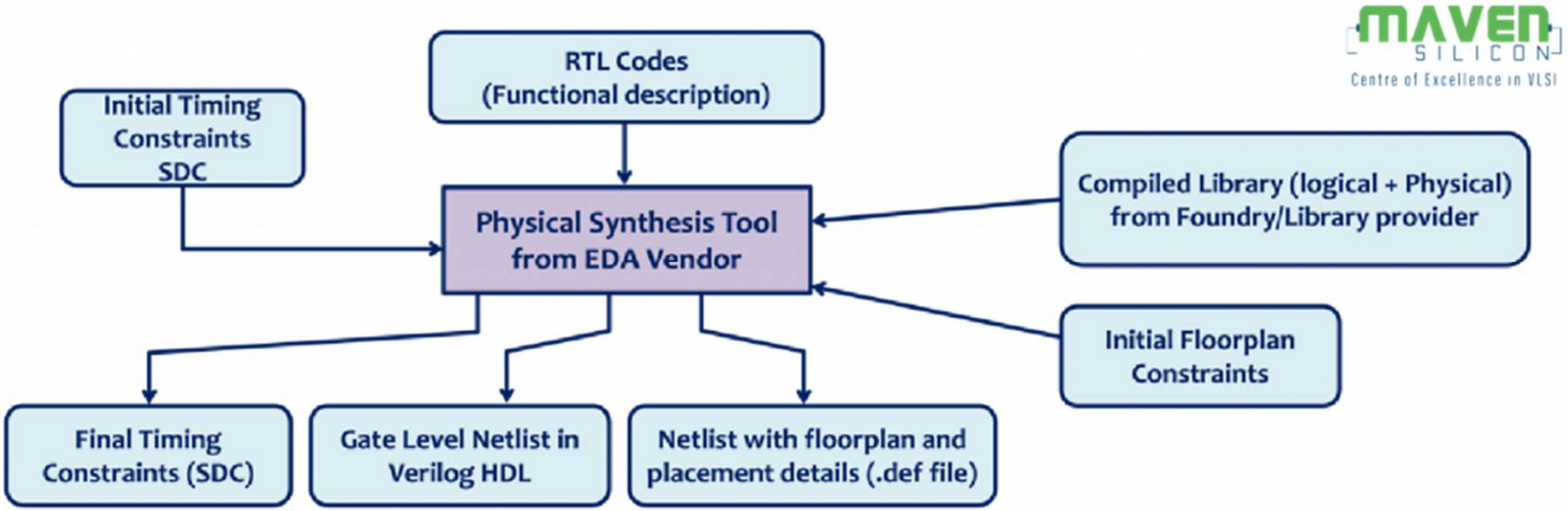
- Synthesis includes three stages: translating, mapping, and optimization
- Synthesis considers the process and operating condition to make sure the timing is closure and the area/power is minimized
- Physical cell positions and physical wire typology are usually not considered





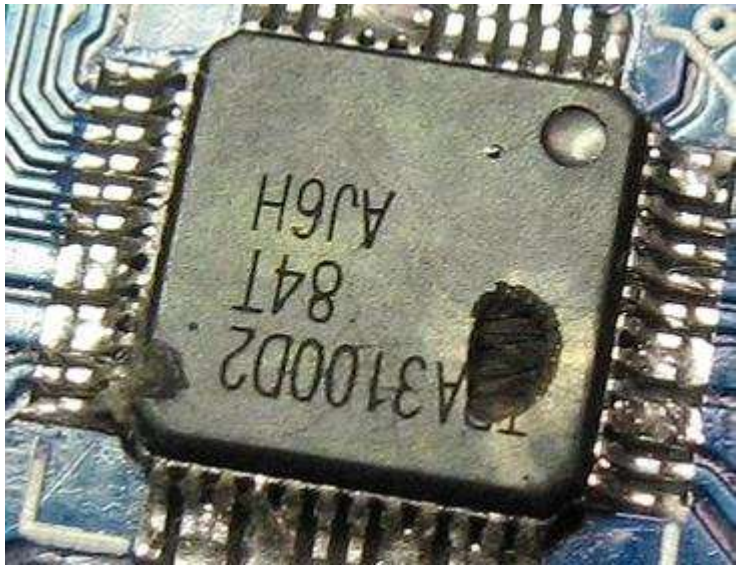
Physical Synthesis

- Physical synthesis considers the physical placement and routing typology
  - More actual wire load and timing
  - Improved routability and reduced DRV
  - Predictable and convergent design flow

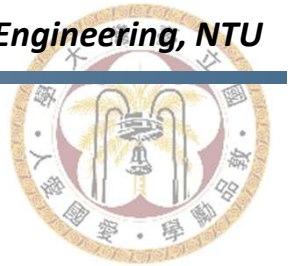


Low-Power Design

- Higher power causes higher temperature, which may lead to permanent damage to the chip and the device
- Portable devices require low power



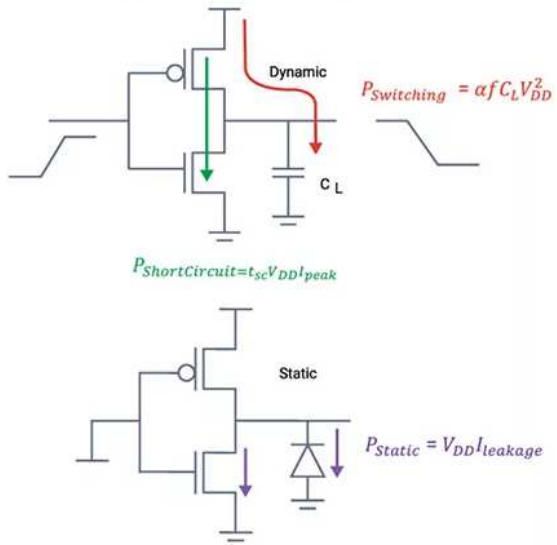




Low-Power Design

- **Dynamic power**
  - Reducing supply voltage
  - Reducing the switching rate of the standard cell
- **Static power**
  - Reducing supply voltage
  - Increasing the threshold voltage of the standard cell

$$P_{Total} = \alpha f C_L V_{DD}^2 + t_{sc} V_{DD} I_{peak} + V_{DD} I_{leakage}$$

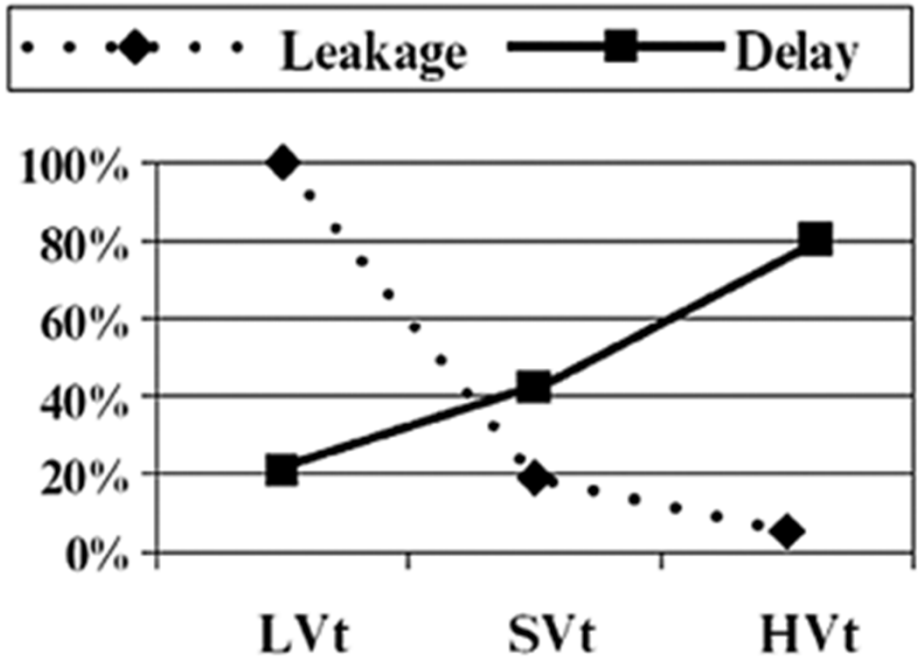


α = activity factor (0 to 1)
f = frequency
t_{sc} = transition time
C_L = capacitive load
V_{DD} = supply voltage
$I_{leakage}$ = leakage current
I_{peak} = peak current

Source: Synopsys

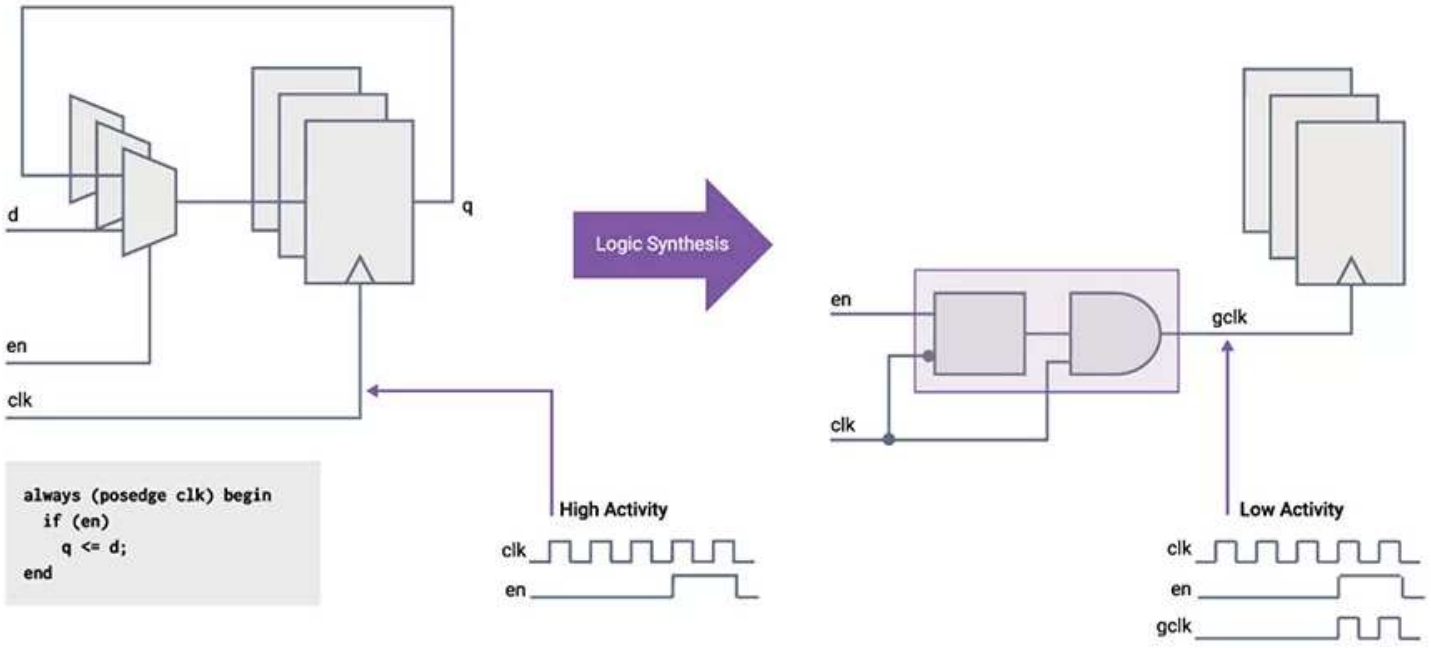
Multi Vt Cell

- High Vt cell can reduce the static current
- Transition of the high Vt cell may be slow
  - May violate setup time requirement
  - May violate max transition requirement



Clock Gating

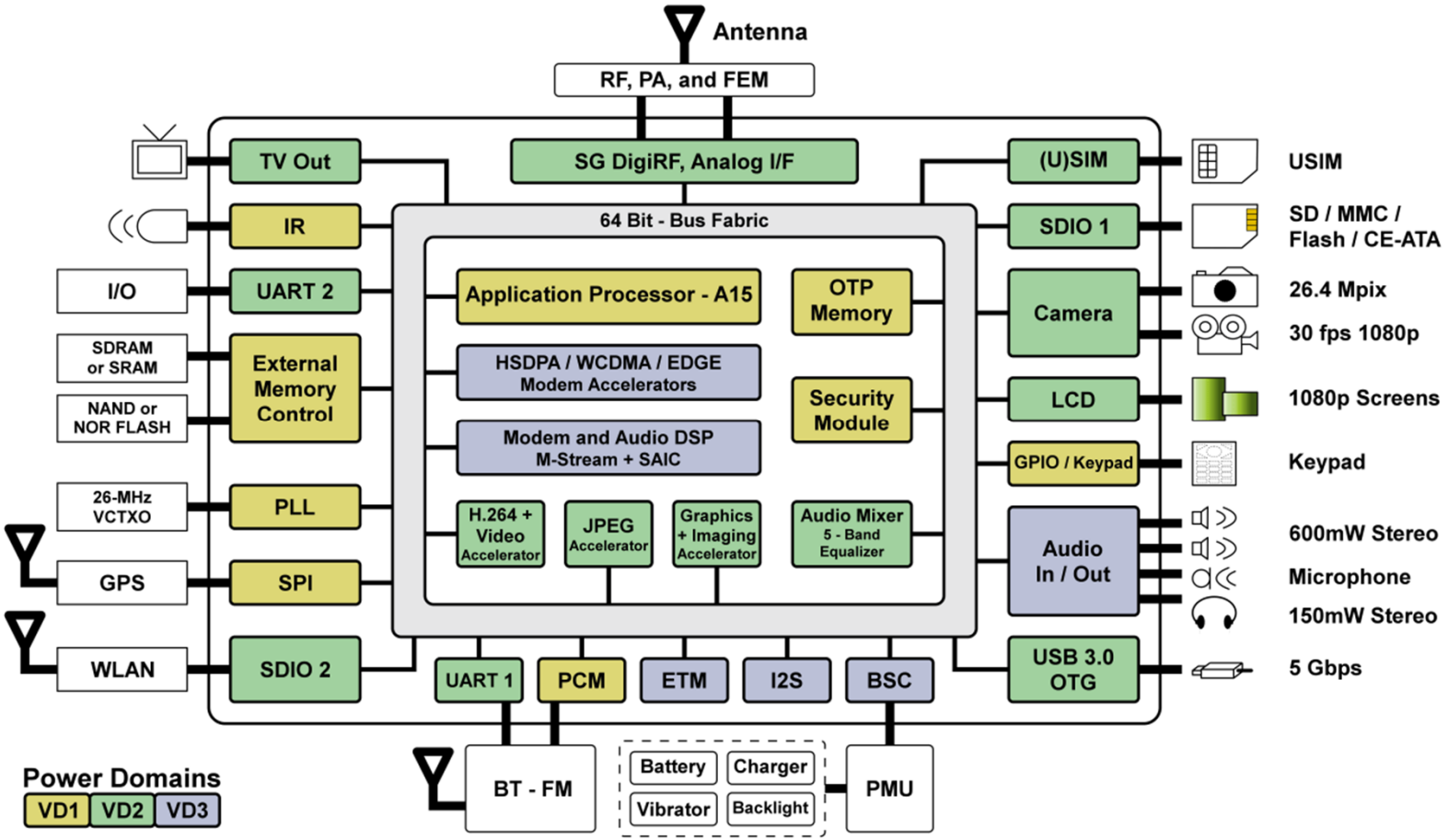
- Clock gating reduces the overall switching activity of the clock net
- Reducing the switching activity -> reducing the dynamic power
- Clock gating can be inserted automatically by synthesis tools



Source: Synopsys

Multiple Clock Domain

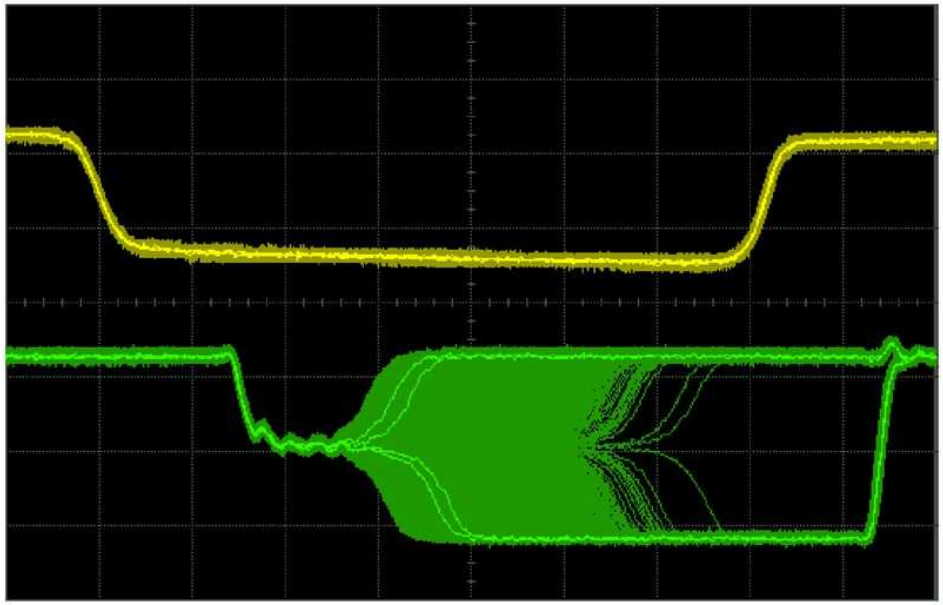
- There may be multiple clock domains in a system



Clock Domain Crossing

- If a signal crosses different clock domains, the timing requirement may be violated, and the DFF may stay in metastability state

Real Metastable Example

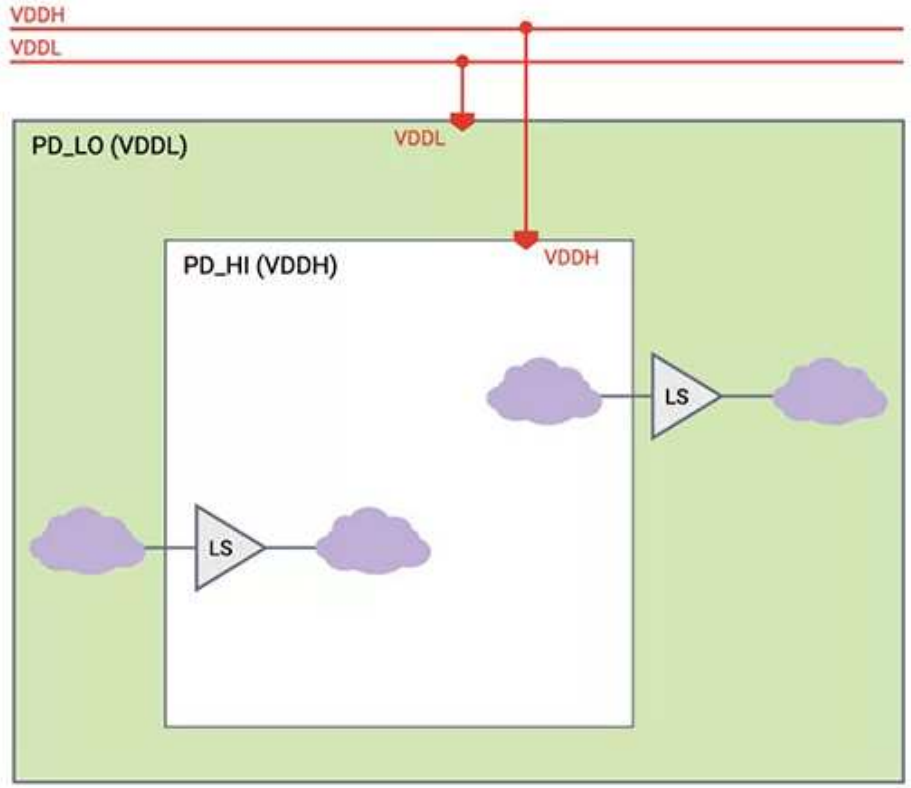


Picture taken from W. J. Dally, Lecture notes for EE108A, Lecture 13: Metastability and Synchronization Failure (ow When Good Flip-Flops go Bad) 11/9/2005.

7

Multi-Voltage Design

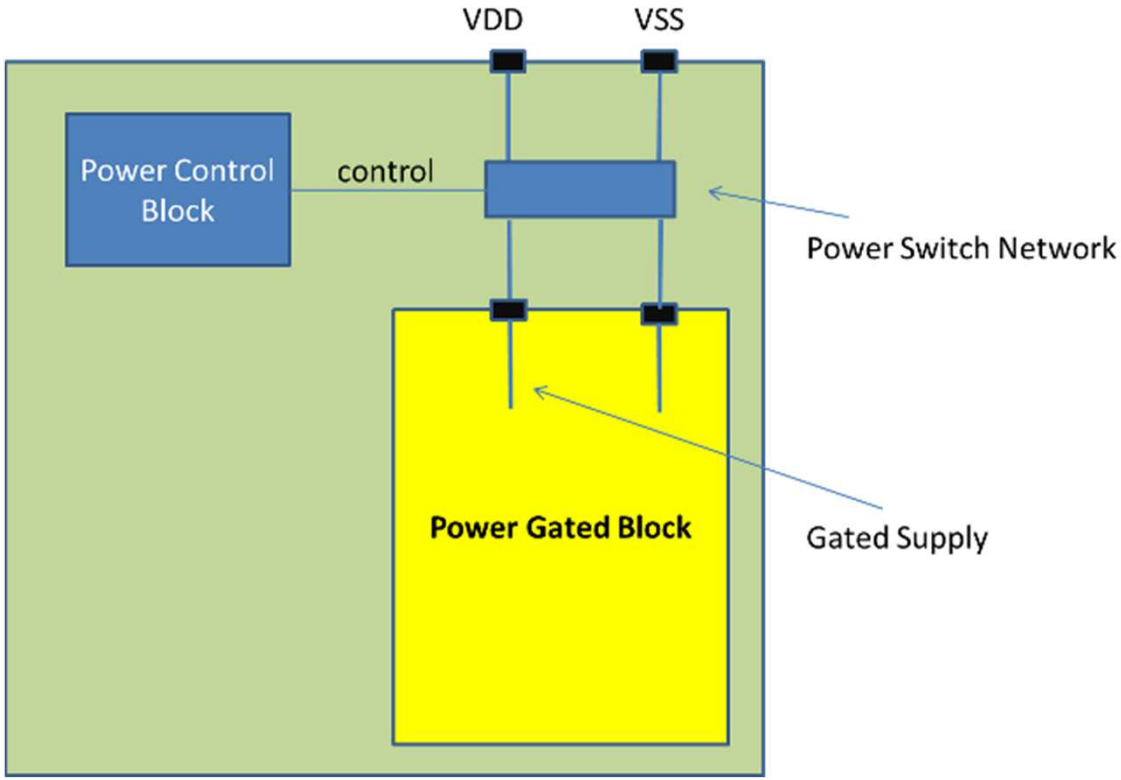
- Different modules may require different voltages
  - High voltage for sequential operating blocks or I/O controllers
  - Low voltage for highly parallel blocks to save power

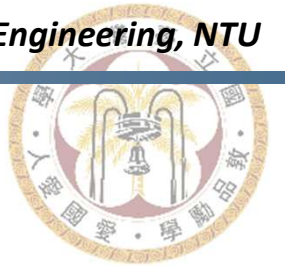


Source: Synopsys

Power Gating

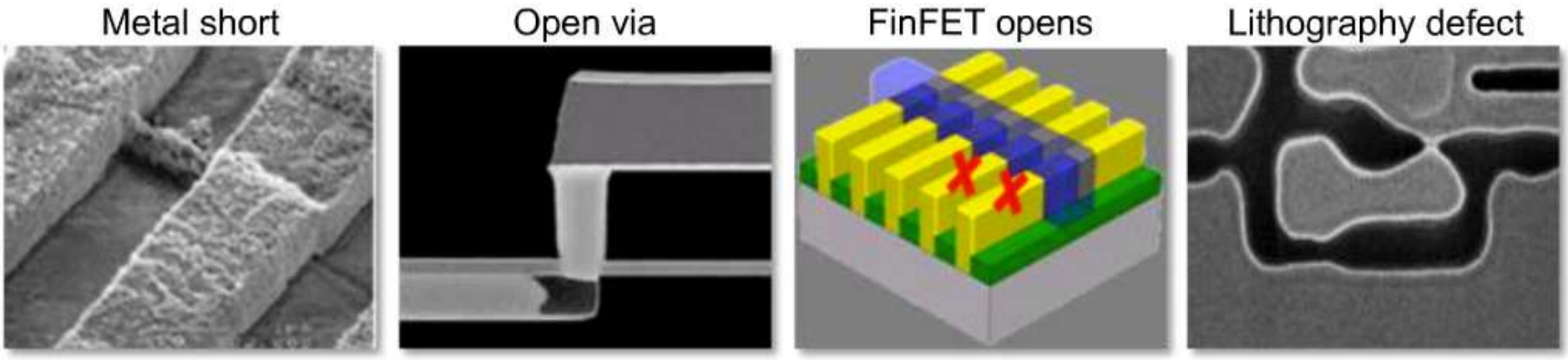
- Power gating is used to cut off the power or ground supply to enable a deep sleep for blocks
- Reduces significant static power





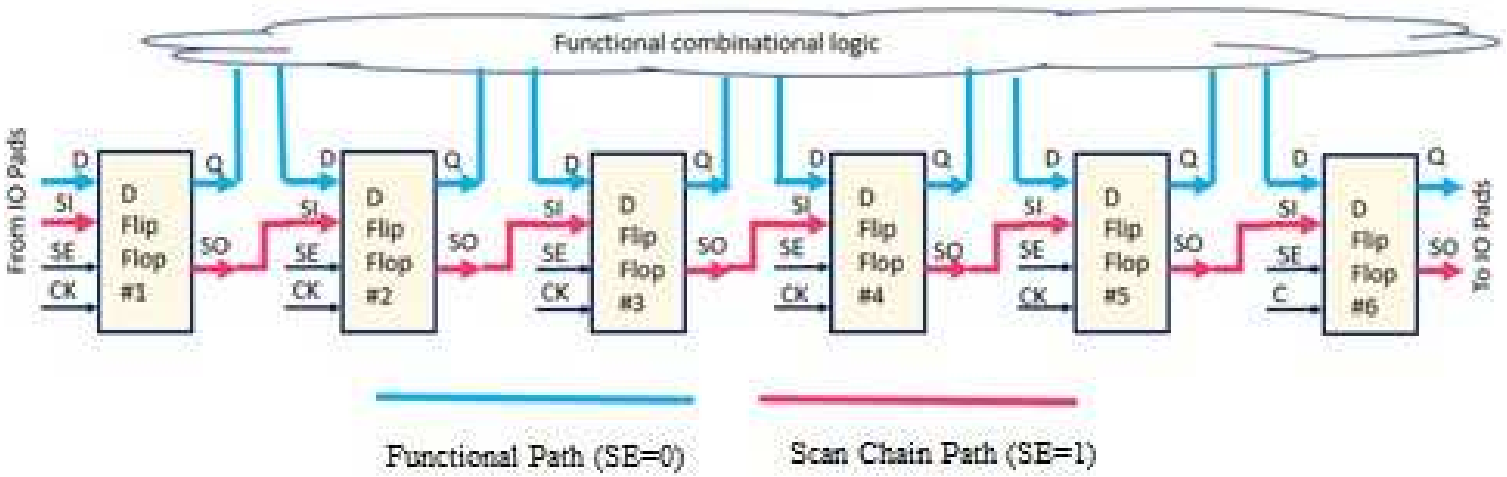
Manufacturing Defects

- A on-chip flaw introduced during fabrication or packaging of an individual ASIC that makes the device malfunction
- Testing is needed to ensure ASIC's correctness and reliability

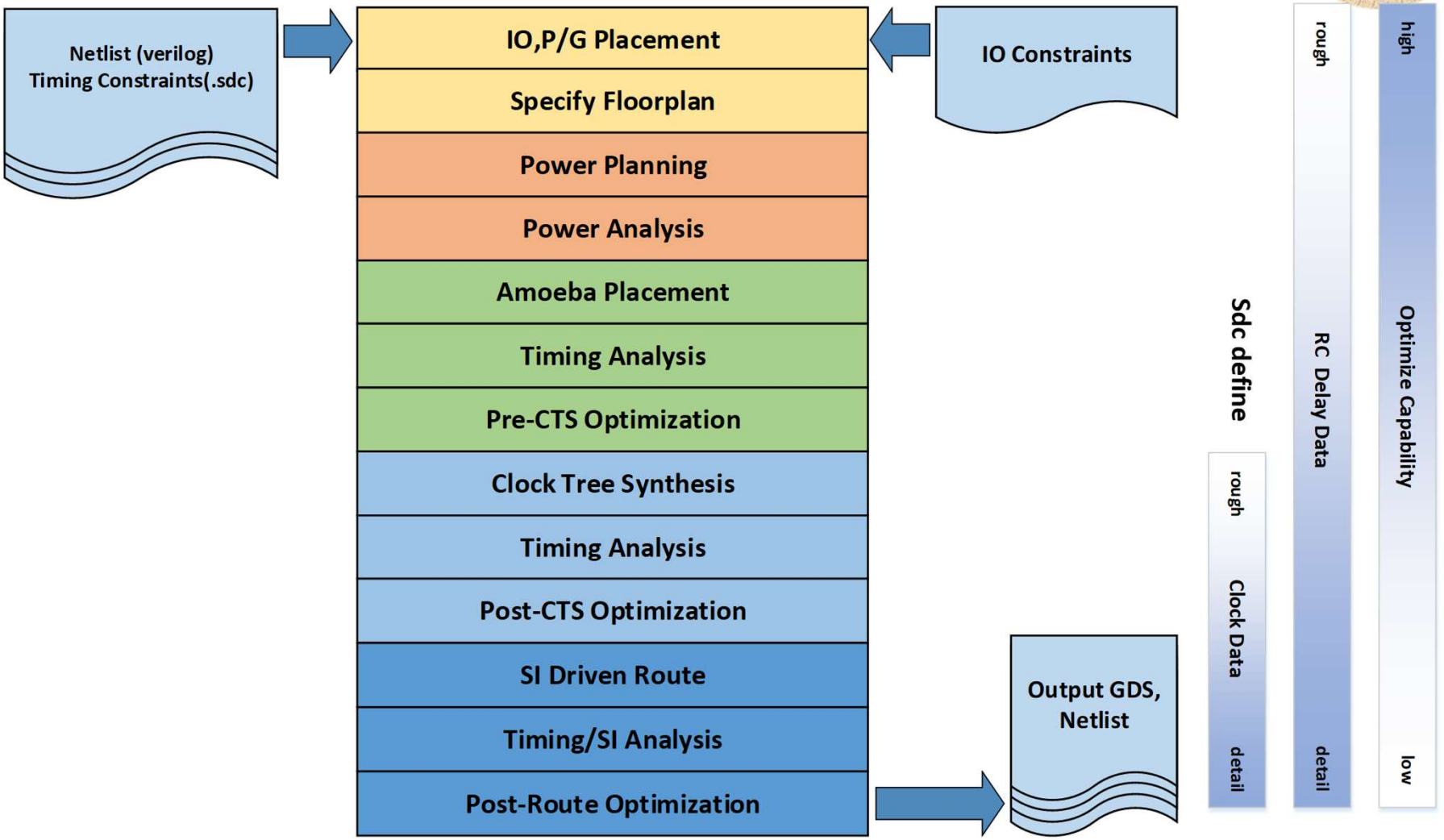


Design for Testability (DFT)

- Benefits of DFT: ensuring quality and cost efficiency
- The test can be done by build-in-self-test circuit and probing machine
- Build-in-self-test circuit
  - Scan chain, boundary scan, memory BIST



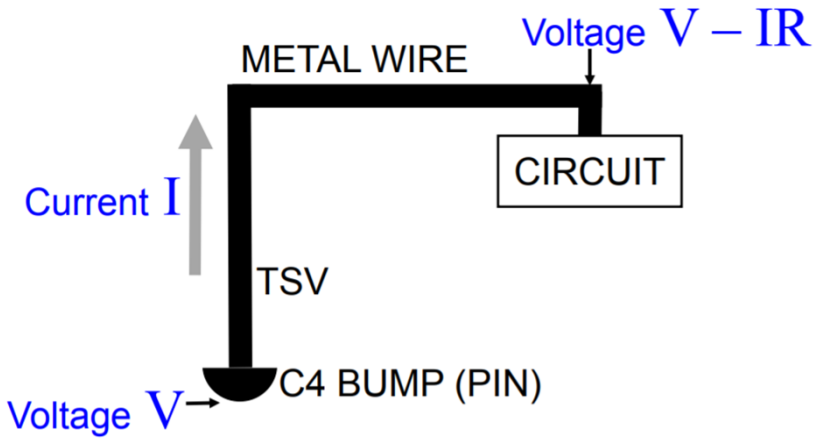
Automatic Place and Route

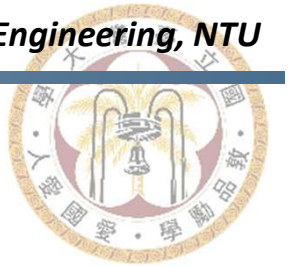


IR Drop Analysis

- The timing of the critical path may not meet the requirement due to the voltage drop of the power supply network
- The IR drop of the power supply network should be considered

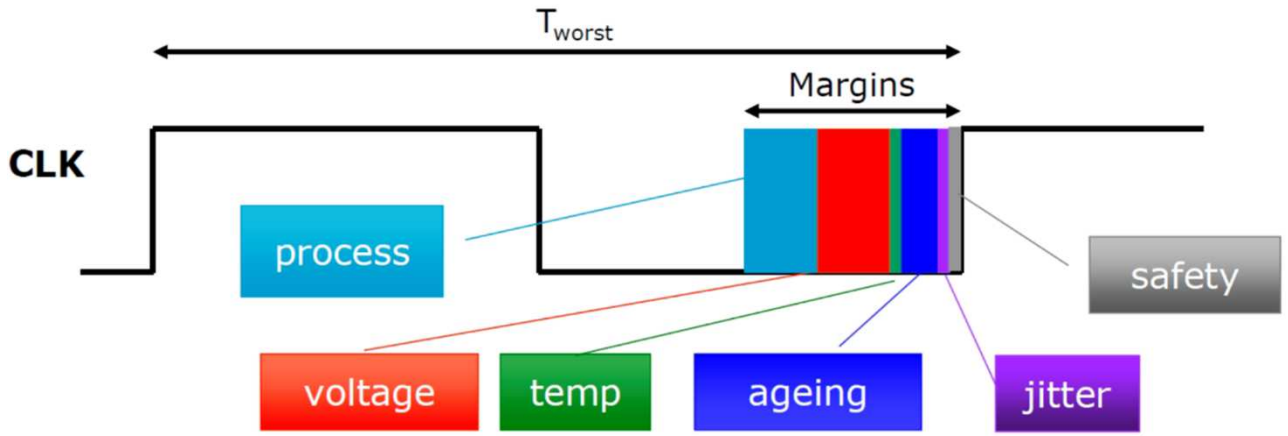






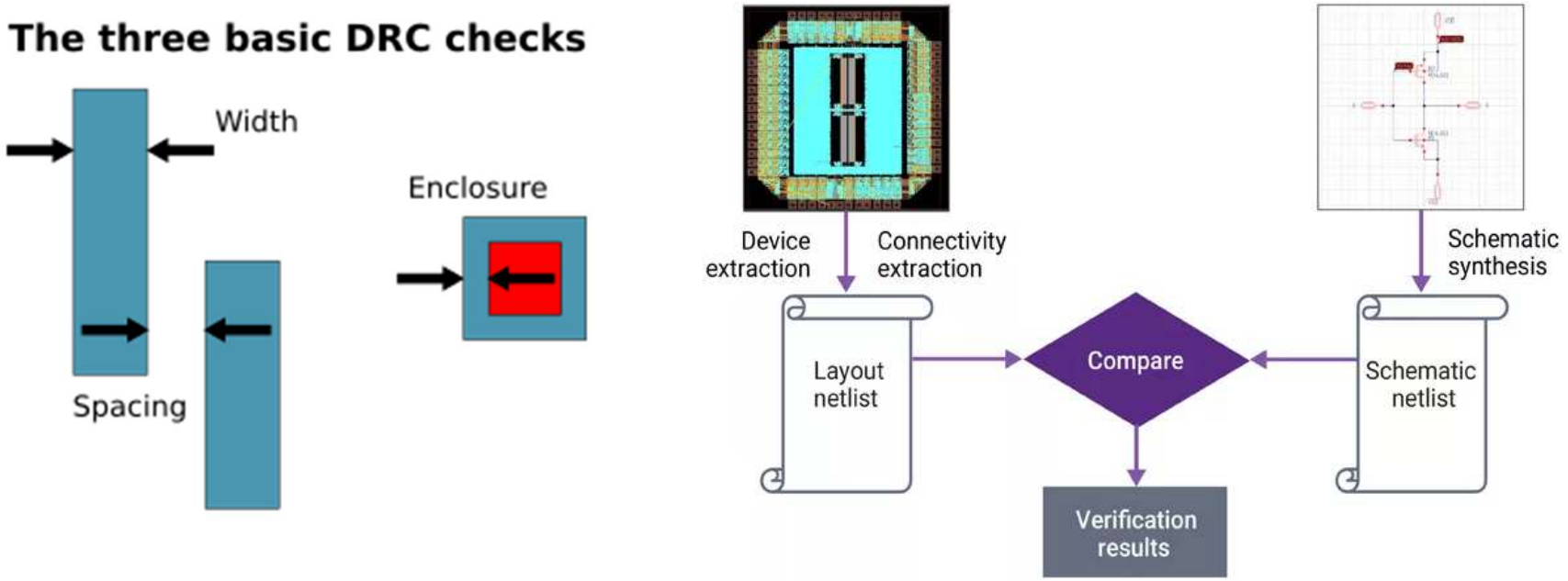
Timing Sign-Off

- **Timing signoff** is a challenging problem to advanced-node IC design
- Margin for the worst-case variation conditions limits F_{max} and V_{min}
- Related constraints to improve productivity
  - On-chip variation (OCV), Advanced OCV, Parametric OCV, Momentum-based OCV
  - LVF (Liberty with variation adjustment) flow
  - Clock uncertainty



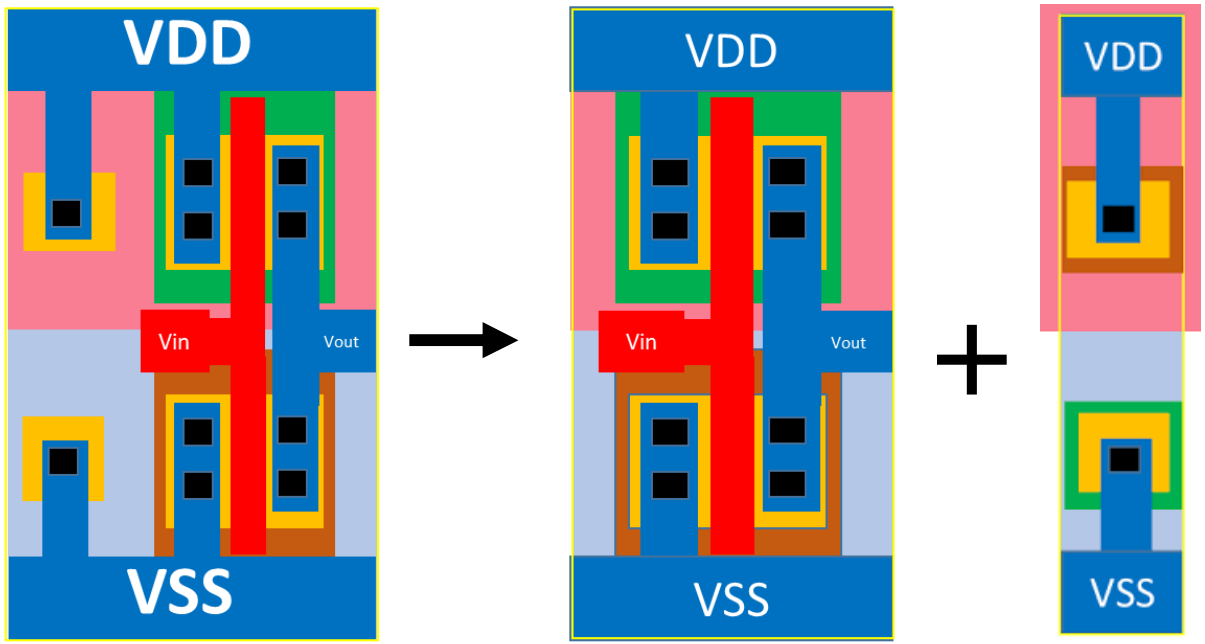
Physical Sign-Off

- Design rule check makes sure the design is producible
- Layout versus schematic makes sure the physical design is the same as the schematic netlist



Design for Manufacturing (DFM)

- DFM ensures designs can be accurately manufactured
- DFM also improves functional yield, parametric yield, and reliability
- Some physical-only cells are added
  - Boundary cell
  - Tap well



Traditional Standard Cell layout

Tapless Standard Cell layout